

and change RMSD (PoseBusters.md, §§2.2–2.6, 3.2–3.3; Supplement §§S3–S8). PoseBench further offers a post-September-2021 filtered PoseBusters subset for newer co-folding models and DockGen-E for novel ECOD pockets (PoseBench.md, §5.2; Appendix D).

Proposal. Use **holo receptors** deliberately in experiment 1. This removes the largest avoidable confound: DiffDock-L holds receptor coordinates rigid while a quantum score can only judge the supplied geometry. With apo/predicted receptors, a correct ligand may clash because side chains/backbone are wrong, and no frozen-pose scorer can separate ranker error from receptor error. The claim must therefore be “cognate holo, fixed-receptor reranking.” Repeat on apo/flexible inputs only after a signal exists.

7. Experiment 1: an executable, falsifiable pilot

7.1 Revised hypotheses and estimands

The original H1 mixes two questions. Pre-register them separately:

H1-physical. Conditional on a frozen DiffDock-L set containing at least one near-native pose, a fixed-geometry SAPT0 interaction score adds within-complex pose-ranking information beyond DiffDock-L confidence and AA-Score when all methods receive identical steric/strain handling.

H1-correlation. Replacing SAPT0 first-order electrostatic/exchange terms by an (8e,8o or smaller) active-space correlated estimate changes rankings in a direction that improves near-native selection; ideal VQE reproduces the same-active-space CASCI correction closely enough not to erase that improvement.

H1-physical can be true while H1-correlation is false. Only H1-correlation tests whether the quantum-representable part is scientifically relevant. A simulator cannot establish computational quantum advantage; it can only establish observable usefulness and circuit/resource feasibility.

Define two estimands:

- **End-to-end top-1 success:** success among all included complexes, where missing oracle coverage is a generator failure.
- **Conditional top-1 reranking success:** success only among complexes with at least one sampled pose at symmetry-corrected RMSD < 2 Å. This is the primary ranker estimand.

Both must be reported. Conditional performance alone cannot be advertised as docking success.

7.2 Dataset and locked cohort

Benchmark. Use the 308-complex **PoseBusters Benchmark set** distributed/evaluated by PoseBusters and PoseBench, not the original PDBind time split. PoseBusters constructed it from recent high-quality complexes and designed explicit validity checks; PoseBench uses the set as its intermediate-difficulty benchmark (PoseBusters.md, §2.6.2 and §3.2; Supplement §§S3–S5; PoseBench.md, §§3.3, 5.2 and Appendix D.2).

Pilot size: 24 complexes. This is enough to reveal gross paired effects and workflow failures while keeping 480 pose evaluations finite; it is not enough for a definitive small effect. The follow-up validation size must be determined from the pilot’s paired discordance/effect estimates, not retroactively by significance.

Selection, performed before viewing any generated-pose RMSD or electronic score:

1. Require a complete holo protein model and one noncovalent primary ligand; exclude covalent binders, unresolved ligand atoms, alternate ligand occupancies that cannot be resolved deterministically, and complexes requiring a metal/cofactor for the primary pilot.
2. Require ligand elements in H/C/N/O/F/P/S/Cl, 12–35 heavy atoms, 1–8 RDKit rotatable bonds, formal charge $-1, 0, +1$, and a closed-shell singlet under the chosen protonation. These restrictions define the scope; they are not claims about general docking.
3. Generate the 20 DiffDock-L candidates without looking at native coordinates. Form the union of complete receptor residues having any heavy atom within 4.5 Å of any candidate ligand heavy atom. Require at most 100 protein heavy atoms in that fixed union and no disjoint region whose covalent capping is chemically ambiguous. This feasibility filter depends on generator geometry, not native RMSD.
4. Bin eligible complexes by neutral/charged ligand and 1–4/5–8 rotatable bonds. Sort PDB IDs within each of four bins by SHA-256 of the literal string `PDB_ID|q-sapt-pilot-v1`; take six from each bin. Freeze IDs,

exclusions, and reasons in a manifest before computing RMSD or scores. If a bin has fewer than six, use the next pre-declared bin in cyclic order and record the deviation.

5. Do not replace a complex because its SCF, VQE, or scorer later fails. Method failure is an outcome. A replacement is allowed only for an objectively corrupted input discovered before any score is inspected, using the next hashed ID and an audit log.

Leakage audit. PoseBusters recency reduces direct DiffDock training leakage but does not prove domain novelty. For every retained complex, record PDB release date, maximum receptor sequence identity and ligand similarity to DiffDock/DiffDock-L training data where manifests permit, and whether the ECOD domain occurs in training. Report subgroup results; do not filter after observing accuracy. Holo coordinates and crystallographic waters can reveal pocket geometry, so the claim is explicitly cognate-holo reranking—not blind apo generalization.

Scientific validity of filtering. Publish the full flow diagram: 308 initial, count/reason at every gate, 24 selected IDs, and descriptor distributions versus all eligible complexes. Conclusions apply only to small-to-medium, closed-shell, nonmetallated ligands and computationally bounded pockets. A later metal/multireference stratum is necessary if the active-space correction is hypothesized to matter most there.

7.3 Pose generation and frozen-candidate contract

1. Use one pinned **DiffDock-L** checkpoint and repository commit. Record whether its confidence architecture is the original or bootstrapping variant.
2. Remove the crystallographic ligand and every feature derived from its coordinates before inference. Supply the holo receptor coordinates and ligand graph/SMILES; do not supply a pocket centre or native conformer.
3. Generate $N = 20$ poses per complex in one prescribed run, with seeds `base_seed + 0...19` (or the repository’s equivalent batched seeds). Preserve the RDKit ETKDG seed conformer, raw trajectory metadata, raw pose, and DiffDock confidence.
4. Never insert the native pose or a native-perturbed pose. Oracle coverage must come from the generator.
5. Primary scores operate on **raw** DiffDock coordinates after adding hydrogens in the fixed preparation pipeline. Do not energy-minimize before primary ranking: relaxation changes the candidate, may collapse several candidates to one basin, and can conceal invalid generation.
6. Secondary sensitivity analysis applies the identical restrained protocol to every candidate: protein heavy atoms fixed; ligand heavy-atom harmonic restraint weak enough to resolve hydrogen contacts but strong enough to report displacement; fixed iteration limit; score both before and after; recompute RMSD and PoseBusters. Never minimize only the quantum-scored arm.

Archive a single candidate table keyed by (complex_id, pose_id, generation_seed). Every scorer reads it and may not change coordinates. This makes comparisons paired and prevents silent scorer-specific docking.

7.4 Fixed quantum-region construction without crystal-pose leakage

This protocol is a **proposal** and must be frozen before RMSD evaluation.

Atoms and boundary

- **Ligand monomer B:** **the complete ligand**, with the same bond order, protonation, charge, and heavy-atom geometry for every scorer.
- **Protein monomer A:** the **union of complete residues within 4.5 Å of any of the 20 candidate poses**, not the native ligand. Use the same atom set for all 20 poses. Include one adjacent residue on each side of a retained contiguous peptide segment where required for **chemically sensible caps**.
- **Cut peptide bonds only.** Cap exposed N/C termini with ACE/NME in a deterministic internal-coordinate construction; optimize cap hydrogens only with all original heavy atoms fixed. Reject, before scoring, regions needing arbitrary cuts through a cofactor or covalent network.
- Exclude crystallographic waters in the primary analysis to avoid inconsistent occupancy. A predeclared secondary analysis can add waters within 3.5 Å of any candidate and fixed by receptor hydrogen-bond criteria, again as one common set across all poses.

The union rule may include atoms irrelevant to a particular pose, but it preserves comparable Hamiltonian composition. A per-pose nearest-residue cutout would make energies incomparable and can reward poses merely for producing a smaller region.

Protonation, charge, and spin

- Add missing receptor atoms once. Assign protonation at pH 7.4 with one pinned tool/version; manually adjudicate only a predeclared list of histidine, Asp/Glu, Lys, Cys, and termini rules while blind to RMSD/scores. Store all decisions.
- Standardize the ligand protonation/tautomer from its benchmark chemical component and input SMILES. Do not enumerate states in primary analysis. Flag ambiguous cases before cohort locking; later state enumeration is a separate experiment.
- Require integral total charges for both monomers and a closed-shell global reference. Active-space N_α, N_β and multiplicity are fixed per complex and identical across poses.

Basis, orbitals, and active space

- Use 6-31G for the primary feasibility run because it matches the large KDM5A precedent and keeps the classical integral problem bounded (SAPT_VQE_PL.md, §II.C and SI §III). This is not a production-quality dispersion basis. On four smallest complexes, repeat SAPT0 with jun-cc-pVDZ or the largest supported SAPT basis to quantify basis-induced rank changes.
- Compute one RHF protein-monomer reference in a **monomer-centred** basis. Localize occupied orbitals (Pipek–Mezey or IAO, chosen once). Treat the ligand at RHF for every unique internal conformer; rigidly related candidates reuse its density exactly up to coordinate transformation.
- Select the protein active space without the native pose. Build an AVAS candidate space from valence orbitals on chemically nontrivial protein atoms that contact at least 25% of the candidate ensemble (polar/charged/aromatic atoms ranked by contact frequency, with deterministic tie-breaking). If larger than 8 spatial orbitals, use a loosely converged SHCI/CASSCF natural-orbital calculation on the *isolated fixed protein monomer* and retain an (8e,8o) space by a frozen occupation rule. If electron count requires it, use (6e,6o); never choose the size after seeing ranking performance.
- Track and report natural occupations. If every occupation is effectively 0 or 2 and CASCI–RHF corrections are negligible, that is evidence against H1-correlation, not permission to switch complexes.
- Map 8 spatial active orbitals to 16 spin-orbital qubits (or 6 to 12) by Jordan–Wigner and use the paper’s shallow number/spin-preserving 1- μ CJ ansatz first. Optimize statevector VQE from RHF; compare total energy and every required RDM contraction to CASCI.

Remainder of protein and solvent

- Assign the receptor outside the SAPT cutout fixed AMBER point charges and Lennard–Jones parameters. Do **not** silently put those charges into only one monomer Hamiltonian. Primary reporting separates SAPT cutout interaction from a classical ligand-remainder MM term computed identically for all SAPT arms.
- Remove cutout atoms and artificial cap atoms from the remainder interaction list. At severed peptide bonds, apply one frozen link-charge rule—zero the boundary-crossing charge group and redistribute its net charge to the first retained remainder shell, or use an established charge-shift scheme—and verify that total receptor charge is conserved. Because the protein is rigid, omitted cutout–remainder self-energy is pose-constant; ligand–remainder terms are not.
- Define $E_{\text{phys}} = E_{\text{SAPT}}(\text{cutout}) + E_{\text{MM}}(\text{ligand}, \text{remainder})$. Report both components so any gain from the classical environment cannot be attributed to the quantum term.
- Use no continuum solvent in primary ranking. On the four-complex sensitivity subset, add a consistent GB/continuum desolvation estimate to all interaction arms. This tests robustness but still does not yield ΔG_{bind} .

Reuse ledger Per complex, cache one protein geometry, SCF, localized orbitals, active Hamiltonian, optimized VQE parameters, and 1-/2-RDM. Reuse them for all 20 poses. Cache ligand RHF densities by exact internal-coordinate hash; rigid translation/rotation only changes transformed intermonomer integrals. Recompute ligand SCF for a new torsional conformer. Compute pose-specific mixed integrals and classical SAPT contractions for every candidate. Log whether a cache hit was exact, symmetry-equivalent, or merely a warm start.

7.5 Scores and baseline matrix

All scores are higher-is-better only after a documented sign conversion; raw values remain archived.

ID	Score/ranker	Purpose
R0	Uniform random candidate, repeated over 10,000 seeded draws	Chance reference conditional on each candidate set
R1	DiffDock-L confidence	Learned native-pose baseline
R2	Vina score-only and, if stable on fixed coordinates, GNINA CNN pose score	Conventional/learned docking-rescore baseline; no search
R3	AA-Score alone	Closest CompassDock interaction baseline
R4	PoseCheck strain, clash count, and PoseBusters validity	Noninteraction physical controls
R5	SAPT0-like second-order score + fixed remainder MM term	Tests H1-physical
R6	$E_{\text{SAPT0}} + [E_1^{\text{CASCI}} - E_1^{\text{RHF}}]$	Exact active-space correlation control
R7	$E_{\text{SAPT0}} + [E_1^{\text{VQE}(\text{statevector})} - E_1^{\text{RHF}}]$	Ideal quantum-representable estimator; must match R6 within tolerance
R8	Shot-sampled/noise-model R7 on a predeclared subset	Ranking stability under measurement error, not advantage
R9	Full second-order SAPT(CASCI/VQE) on size-qualified four-complex subset	Checks whether the mixed hierarchy changes conclusions

Two composites are predeclared; no weights are fit on 24 test complexes:

1. **Validity-controlled interaction:** poses failing PoseBusters’ intermolecular distance/volume-overlap checks are placed after passing poses; within each tier rank by the interaction score. Strain is reported but is not a hard exclusion.
2. **Confidence augmentation:** average the within-complex percentile ranks of DiffDock confidence and the chosen physical score (equal 0.5/0.5 weight), after the same validity tiering. Repeat equal-weight fusion for AA-Score so the learned-confidence contribution is controlled.

Report pure scores and composites separately. A win caused solely by the validity tier is a PoseCheck/PoseBusters result, not a quantum result.

Q-Score decision. Do not place published Q-Score in the primary matrix because it re-solves an anchor graph and reconstructs a rigid pose rather than scoring the supplied pose. If its code is available, an exploratory appendix may: (i) run its exact classical clique solver, not QAOA; (ii) freeze one global τ without crystal tuning; and (iii) quantify how much the Kabsch output moves from each input. Label any fixed-coordinate interaction sum as a modified SIMG* baseline, not Q-Score.

7.6 Metrics and failure attribution

For every score, report:

- **Oracle coverage:** proportion with $\min_{j \leq 20} \text{RMSD}_{ij} < 2 \text{ \AA}$; additionally top-5/top-10 coverage by the generator’s original order.
- **Top-1 and top-3 success:** symmetry-corrected ligand RMSD $< 2 \text{ \AA}$, both end-to-end and conditional on oracle coverage.
- **Continuous pose quality:** selected-pose RMSD and centroid distance; do not let a few huge errors dominate—report median and full paired distribution.
- **Within-complex rank relation:** Spearman correlation between higher score and negative RMSD for each complex, summarized by median and a complex bootstrap. Correlations are undefined when RMSD/score is constant; report the count rather than impute zero.
- **Pairwise discrimination:** probability that a near-native pose ($< 2 \text{ \AA}$) outranks a clear decoy ($> 4 \text{ \AA}$), using all eligible pairs but cluster-bootstrap by complex. Report intermediate 2–4 $\text{\AA}$ poses separately.

- **Physical validity:** PoseBusters pass rate of selected top-1 and joint success (RMSD <2 Å AND PB-valid). Include ligand strain and clash distributions.
- **Interaction recovery:** if feasible, PoseBench PLIF-WM or ProLIF similarity to the native interaction fingerprint as a secondary metric, never as a score input (PoseBench.md, §§5.3–5.4 and Appendix E).
- **Operational feasibility:** preparation/SCF/VQE/SAPT failure rate, wall time, peak RAM/GPU memory, unique QPU states, circuits, shots, and cache reuse.

Classify every complex-score result:

1. **Generator failure:** no pose below 2 Å exists among 20; no ranker could succeed.
2. **Ranker failure:** a below-2-Å pose exists but the selected pose is not below 2 Å.
3. **Validity conflict:** the lowest-RMSD candidate fails a physical check or the most physical candidate has high RMSD.
4. **Method failure:** the scorer did not return a comparable value under the locked protocol.

7.7 Minimum ablations

Run these in order; stop expensive later items if an earlier gate fails.

1. **Interaction model:** confidence vs AA-Score vs Vina/GNINA vs SAPT0, with no quantum correction.
2. **Quantum correlation:** SAPT0/RHF first order vs CASCI-corrected vs ideal-VQE-corrected on identical orbitals. This is the decisive quantum ablation.
3. **Validity/strain:** every interaction score alone, plus identical clash tier, plus identical clash tier and equal-rank strain feature. This detects gains caused by PoseCheck rather than interaction physics.
4. **Learned confidence:** physical score alone versus fixed 50/50 rank fusion with confidence. This tests replacement versus augmentation.
5. **Region/active space:** on four locked representatives only, 4.5 versus 6.0 Å region where feasible and (6e,6o) versus (8e,8o). Do not search for the best setting on all 24.
6. **Measurement:** exact statevector, finite-shot grouped measurement, and a pinned device-noise model on two complexes with the smallest and largest correlated rank changes. Repeat sampling 100 times to estimate top-1 stability.
7. **Full second order:** on up to four systems whose **entire locked cutout** genuinely fits the implementation’s orbital/memory limit, compare the mixed hierarchy with full SAPT(CASCI/VQE)-ERPA across their poses. If no whole cutout fits, do not quietly shrink the region per pose. One fixed, predeclared ligand–residue microcomplex may be used to verify equations, but it is not a validation of pocket-pose ranking. If full-method rankings differ materially, the hybrid cannot be generalized without the full method.

7.8 Statistical analysis

- Treat the **complex**, not a pose pair, as the independent sampling unit.
- For top-1 success differences, report the paired difference in percentage points, exact McNemar test on discordant complexes, and a 10,000-replicate stratified complex bootstrap confidence interval. Report the discordance table itself.
- For RMSD, per-complex Spearman, pairwise accuracy, and validity, use paired complex bootstrap intervals. A permutation/sign-flip test is suitable for a predeclared continuous paired summary if its exchangeability assumptions hold.
- Use a hierarchical test order: H1-physical first (SAPT0 composite vs confidence control), H1-correlation second (CASCI correction vs SAPT0), VQE fidelity third (VQE vs CASCI). Stop formal testing after a failed gate; label remaining comparisons exploratory. If many baseline contrasts are interpreted, use Holm correction.
- Report effect sizes and intervals even when a p-value is large. With 24 complexes, “not significant” is not equivalence. Define practical equivalence bands before analysis, e.g. ± 5 percentage points in conditional top-1 and ± 0.03 pairwise accuracy.
- Keep all 24 in operational summaries. The conditional analysis excludes only generator failures by definition, not method failures; method failures should be assigned a prespecified worst rank or reported in a parallel complete-case sensitivity analysis.

7.9 Pre-registered success, continuation, and falsification criteria

Feasibility gate: at least 20/24 complexes yield comparable SAPT0 and active-space results for all 20 poses; the 16-qubit statevector/CASCI RDM agreement meets a fixed tolerance; and at least 12 complexes have oracle coverage. Otherwise redesign preparation or increase N , but do not interpret ranking.

Progress to a larger physical-score validation if SAPT0 augmentation has at least four more conditional top-1 wins than losses versus confidence-only, improves pairwise accuracy by at least 0.05, has a positive lower bound on an 80% pilot bootstrap interval, and does not reduce joint PB-valid/top-1 success. These are pilot continuation thresholds, not proof.

Physical score useful, quantum unnecessary if SAPT0 passes that gate but CASCI correction changes conditional top-1 by less than the ± 5 -point equivalence band, changes pairwise accuracy by less than ± 0.03 , and natural occupations/RHF-CASCI differences show weak correlation. This is a scientifically valuable likely result.

Quantum component adds measurable information only if CASCI-corrected SAPT beats SAPT0 by the same predeclared directional criteria, the effect is not driven by one complex or by validity penalties, VQE reproduces CASCI rankings within uncertainty, and the active-space selection is stable. On a simulator, phrase this as “a quantum-representable correlated observable adds information,” not “quantum advantage.”

Falsify reranking for this scope if oracle coverage is adequate but SAPT0 and its correlated variants have no practically positive conditional effect, systematically favor high-strain/invalid decoys, or lose to AA-Score/Vina under identical validity handling. Also stop if region/basis/protonation sensitivity reverses rankings more often than the purported gain.

Hardware go gate: only after a correlated effect exists, grouped shot estimates preserve at least 90% of the CASCI pairwise decisions at an affordable predeclared shot budget, and the QPU result is compared with a noise-matched classical sample. A decorative one-pose circuit is not evidence for docking utility.

8. Future work: how quantum-informed reverse-diffusion guidance could actually be defined

This section is a derivation and cost test, not an experiment-1 commitment.

8.1 Scalar-energy finite differences on $m + 6$ coordinates

At current pose x , let $A_g(\delta, x)$ be DiffDock’s exact action for component g : Cartesian translation, exponential-map rotation around the unweighted ligand centroid, or its RMSD-aligned torsion action. For coordinate basis vector e_a , a central derivative is

$$\frac{\partial E}{\partial q_a} \approx \frac{E(A(+h_a e_a, x)) - E(A(-h_a e_a, x))}{2h_a}.$$

There are $d = m + 6$ coordinates, so a full central gradient costs $2(m + 6)$ energy evaluations per reverse step per trajectory; forward differences cost $m + 7$ including a shared reference but are biased and fragile under noisy energies. Use separate, convergence-tested h for Å translation and radian rotation/torsion. The displaced geometries must use DiffDock’s actions, not arbitrary Cartesian coordinate nudges.

For N samples and T reverse steps,

$$N_{E,\text{central}} = 2NT(m + 6).$$

With $N = 20$, nominal $T = 20$, and $m = 6$, this is 9,600 electronic-energy evaluations (8,640 if the effective 18-step implementation is used). Every “energy evaluation” may itself require VQE optimization and many measurement groups. Activating guidance only for the last four steps still costs 1,920 energies. Monomer-state reuse helps SAPT under rigid motions, but torsion changes, dimer-centred bases, and pose-dependent embeddings erode it.

A ChemGuide-like simultaneous perturbation can estimate a stochastic direction from two oracle calls,

$$\widehat{\nabla E} = \frac{E(A(+hu, x)) - E(A(-hu, x))}{2h} u,$$